

Break down protocol barriers and make industrial digitization more convenient and efficient



## IG902 Series High-performance Edge Computer

· Multiple Access

· Rich Interfaces

· Built-in DSA

· Cloud Management

# 1. Product Overview

IG902 is a high-performance industrial edge gateway that combines rich interfaces, protocol interoperability, and cloud-ready edge computing for IIoT deployments.

Key features:

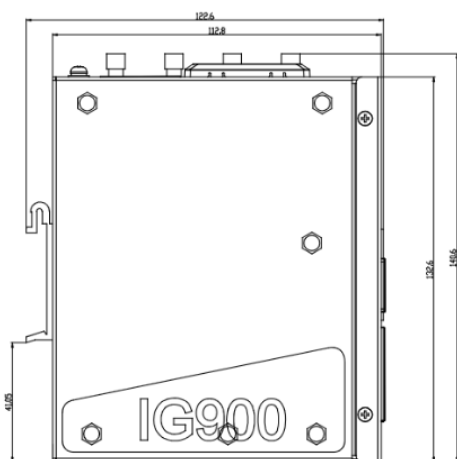
- **High-performance edge compute:** ARM Cortex-A8 1GHz with 1GB DDR3 and 8GB eMMC
- **Rich industrial interfaces:** GE ports, serial, optional DI/DO/relay, USB, MicroSD, optional Wi-Fi/GPS
- **Reliable industrial networking:** Wired/cellular/Wi-Fi backup, dual SIM failover, watchdog, link self-healing
- **Open development:** Python and Docker-based secondary development platform
- **Cloud O&M readiness:** DeviceLive + DSA for remote management and low-code data integration

## Core Technical Specifications

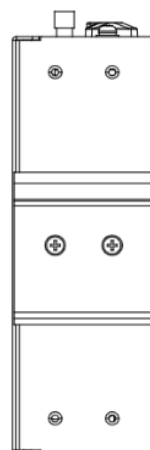
| Specification Item    | Value                                                                                            |
|-----------------------|--------------------------------------------------------------------------------------------------|
| Cellular Network      | LTE Cat4/Cat6 (model dependent)                                                                  |
| Network Features      | APN, VPDN, CHAP/PAP/MS-CHAP/MS-CHAPV2; DHCP Server/Relay/Client; DNS Relay; DDNS; static routing |
| Security              | SPI firewall, ACL, NAT/PAT/DMZ, AAA (Local/Radius/Tacacs+/LDAP), IPSec/GRE/L2TP/OpenVPN/CA       |
| Cloud Management      | DeviceLive remote configuration, upgrades, and operations                                        |
| Secondary Development | Python and Docker secondary development                                                          |

| Specification Item                   | Value                                                                                 |
|--------------------------------------|---------------------------------------------------------------------------------------|
| Data Acquisition Protocols           | Modbus RTU/TCP, EtherNet/IP, OPC UA, IEC101/104, DNP3.0, BACnet, CNC                  |
| Processor and Memory                 | ARM Cortex-A8 @1GHz, 1GB DDR3                                                         |
| Storage                              | 8GB eMMC, MicroSD up to 32GB                                                          |
| Ethernet Ports                       | 2 × 10/100/1000Mbps                                                                   |
| Serial and I/O                       | 1×RS232/RS485 + 1×RS485 (2×RS485 on some models); optional 4DI+4DO or 4DI+3DO+1 relay |
| Power Supply                         | 12~48V DC, reverse polarity and overcurrent protection                                |
| Operating Temperature and Protection | -25~70°C, IP30                                                                        |

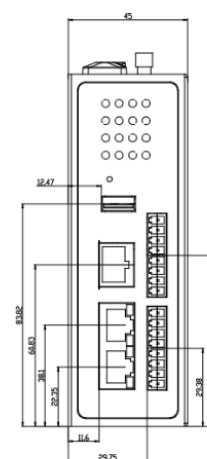
## 2. Product Dimensions & PIN Definition



Front View



Side View



Interface Diagram

### Note:

1. All dimensions are in millimeters (mm).
2. All dimensions are approximate and for reference only.
3. Dimensioned drawings are not intended for machining.

- 4. Dimensions are subject to part and manufacturing tolerances.
- 5. Specifications may change without prior notice.

## 7 PIN Definition

| PIN | Definition | Description                          |
|-----|------------|--------------------------------------|
| 1   | V+         | Positive electrode                   |
| 2   | V-         | Negative electrode                   |
| 3   | TXD/A      | Serial RS232 send / Serial RS485+    |
| 4   | RXD/B      | Serial RS232 receive / Serial RS485- |
| 5   | GND        | Serial RS232 signal ground           |
| 6   | A          | Serial RS485+                        |
| 7   | B          | Serial RS485-                        |

## I/O Definition

| PIN | Definition | Description                |
|-----|------------|----------------------------|
| 1   | PCOM       | Dry contact access point   |
| 2   | DGND       | Dry contact ground point   |
| 3   | DICOM      | Input common port          |
| 4   | DI0        | Digital/pulse input port 0 |
| 5   | DI1        | Digital/pulse input port 1 |
| 6   | DI2        | Digital/pulse input port 2 |
| 7   | DI3        | Digital/pulse input port 3 |
| 8   | NC         | None                       |

|    |                  |                                                                      |
|----|------------------|----------------------------------------------------------------------|
| 9  | DO0              | Digital/pulse output port 0                                          |
| 10 | DGND             | Ground                                                               |
| 11 | DO1              | Digital/pulse output port 1                                          |
| 12 | DGND             | Ground                                                               |
| 13 | DO2              | Digital/pulse output port 2                                          |
| 14 | DGND             | Ground                                                               |
| 15 | DO3/Relay Output | DO3: Digital/pulse output port 3;<br>Relay output: 1A 250VAC / 30VDC |
| 16 | DGND             | Ground                                                               |

**Notes:****DI input specification:**

- Dry-contact status "1": closed
- Dry-contact status "0": disconnected
- Wet-contact status "1": +10 ~ +30V / -30 ~ -10VDC
- Wet-contact status "0": 0 ~ +3V / -3 ~ 0V
- Isolation: 3000VDC
- Pulse signal counter supported, up to 100Hz pulse signal

**DO output specification:**

- Isolation: 3000VDC

## 3. Hardware Specifications

| Category/Parameter                 | Specification                                                                              |
|------------------------------------|--------------------------------------------------------------------------------------------|
| <b>CPU and Storage</b>             |                                                                                            |
| CPU                                | ARM Cortex-A8 @1GHz                                                                        |
| RAM                                | IG902-B: 512MB DDR3<br>IG902-H: 1GB DDR3 RAM                                               |
| Flash                              | 8GB eMMC                                                                                   |
| <b>Connectivity and Interfaces</b> |                                                                                            |
| Ethernet Ports                     | 2×10/100/1000Mbps Ethernet ports (WAN/LAN or 2×LAN)                                        |
| I/O Ports                          | None / 4×DI + 3×DO + 1×Relay output DO or digital/pulse output DO                          |
| Serial Ports                       | 1×RS232/RS485 + 1×RS485                                                                    |
| SIM Card Slot                      | 1.8V/3V, 2×drawer-type slot                                                                |
| LED Indicators                     | POWER, STATUS, WARN, ERROR, MODEM, SIM1, SIM2, TF, PYTHON, USER1, USER2, WIFI, GPS, SIGNAL |
| Console Port                       | 1×console RS232 (RJ45)                                                                     |
| USB Port                           | 1×USB 2.0 port                                                                             |
| TF                                 | MicroSD expansion up to 32GB                                                               |
| Wi-Fi(optional)                    | 2.4G/5G Wi-Fi (802.11 ac/a/b/g/n)                                                          |
| GNSS(optional)                     | GPS and BeiDou                                                                             |
| Reset Button                       | Pinhole button                                                                             |

| Category/Parameter                      | Specification                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Power and Power Consumption</b>      |                                                                                                      |
| Power Input                             | 12~48V DC input                                                                                      |
| Power Terminal                          | Unpluggable industrial terminal                                                                      |
| Reverse Polarity/Overcurrent Protection | Supported                                                                                            |
| <b>Mechanical Specifications</b>        |                                                                                                      |
| Mounting Method                         | DIN-rail/wall mounting                                                                               |
| Protection Rating                       | IP30                                                                                                 |
| Housing and Cooling                     | Metal housing, fanless                                                                               |
| RTC (Optional)                          | Embedded RTC powered by super capacitor                                                              |
| <b>Environment and Certifications</b>   |                                                                                                      |
| Storage Temperature                     | -40~85°C                                                                                             |
| Operating Temperature                   | -25~70°C                                                                                             |
| Ambient Humidity                        | 5~95% RH non-condensing                                                                              |
| Physical Characteristics                | IEC60068-2-27 shock resistance<br>IEC60068-2-6 vibration resistance<br>IEC60068-2-32 drop resistance |

| Category/Parameter | Specification                                                                                                                                                                                                                                                                                                                                                     |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EMC Standard       | EN61000-4-2, level 3, Static<br>EN61000-4-3, level 3, Radiation Electric Field<br>EN61000-4-4, level 3, Pulsed Electric Field<br>EN61000-4-5, level 3, Surge<br>EN61000-4-6, level 3, Conducted Disturbance Immunity<br>EN61000-4-8, Power Frequency Field Resistance,<br>horizontal / vertical 400A/m (>level 3)<br>EN61000-4-12, level 3, Shock Wave Resistance |
| Certifications     | CE,FCC, PTCRB, RCM, IC, IMDA, AT&T, MIC&JATE, MSIP, EAC,ANATEL, UKCA                                                                                                                                                                                                                                                                                              |

## 4. Software Specifications

| Category/Parameter    | Specification                                                                                          |
|-----------------------|--------------------------------------------------------------------------------------------------------|
| Operating System      | Custom Linux                                                                                           |
| Network Features      |                                                                                                        |
| Network Access        | APN, VPDN                                                                                              |
| Access Authentication | CHAP/PAP/MS-CHAP/MS-CHAPV2                                                                             |
| Network Type          | LTE, WCDMA(HSPA+), EDGE, GPRS, CDMA                                                                    |
| LAN Protocols         | ARP, EtherNet                                                                                          |
| IP Applications       | Ping, Traceroute, DHCP Server/Relay/Client, DNS Relay, DDNS, Telnet, SSH, HTTP, HTTPS, TFTP, FTP, SFTP |
| IP Routing            | Static routing                                                                                         |
| Security              |                                                                                                        |
| User Management       | Multi-level users                                                                                      |

| Category/Parameter                                        | Specification                                                                                          |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Network Security                                          | SPI firewall, anti-DoS attack, multicast/ping filter, ACL, NAT, PAT, DMZ, port mapping, virtual server |
| Data Security                                             | IPSec VPN, GRE, L2TP, OpenVPN, CA                                                                      |
| CA Certificates                                           | Supported (may auto apply)                                                                             |
| AAA<br>(Authentication,<br>Authorization,<br>Accounting)  | Local/Radius/Tacacs+/LDAP                                                                              |
| <b>Reliability</b>                                        |                                                                                                        |
| Link Detection                                            | Heartbeat packet detection, auto-recovery of disconnection                                             |
| Embedded Watchdog                                         | Device self-diagnosis, auto-recovery from operation faults                                             |
| Backup Mechanism                                          | VRRP, interface backup                                                                                 |
| Dual-SIM Switching                                        | Dual-SIM backup                                                                                        |
| <b>WLAN (Optional)</b>                                    |                                                                                                        |
| WLAN Standard                                             | IEEE 802.11 ac/a/b/g/n                                                                                 |
| WLAN Security                                             | Open System, Shared Key, WPA/WPA2, WEP/TKIP/AES encryption                                             |
| WLAN Mode                                                 | AP, Client modes                                                                                       |
| <b>Open Platform and Data Acquisition Protocols (DSA)</b> |                                                                                                        |
| Python Secondary Development                              | Secondary development platform with Python and Docker                                                  |

| Category/Parameter        | Specification                                                                                                                                                                |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IoT Platform              | Microsoft Azure, Amazon AWS, Alibaba Cloud, etc.                                                                                                                             |
| Industrial Protocols      | Modbus RTU Master/Slave, Modbus TCP Master/Slave, EtherNet/IP, ISO on TCP, OPC UA Client/Server, Mitsubishi MC 3C/3E/3C OverTCP, Mitsubishi CPU Port, FINSUDP, HostLink, PPI |
| Electricity Protocols     | DLT645-2007, IEC101/104, DNP3.0                                                                                                                                              |
| Other Protocols           | BACnet, CNC                                                                                                                                                                  |
| <b>Network Management</b> |                                                                                                                                                                              |
| Configuration Methods     | Local/remote HTTP/HTTPS/Telnet/SSH config                                                                                                                                    |
| Upgrade Methods           | WEB/DeviceLive/TFTP/FTP/SFTP upgrades                                                                                                                                        |
| Log                       | Local or remote log export, power-down log saving                                                                                                                            |
| Remote Management         | DeviceLive-based remote access and remote batch device management                                                                                                            |
| Network Diagnostics       | Ping, Traceroute, Sniffer (network packet capture tool)                                                                                                                      |

## 5. Ordering Information

### Model Rule

**Model code:** IG902-<B/H>-<WMNN>-<IO/RIO/NA>-<DW/NA>-<G/NA>

<B/H>: Product version (H shown in this release)

<WMNN>: Cellular Type & Frequency Band

<IO/RIO/NA>: I/O option ( **I0** =4DI+4DO, **RI0** =4DI+3DO+Relay, **NA** =No extended I/O)

<DW/NA>: Wi-Fi option ( **DW** =Wi-Fi enabled)

&lt;G/NA&gt;: GPS option

## Model List

| Model                      | Version     | Region                          | <WMNN>: Cellular Type & Band                                                                                                     | Serial Port          | <IO/RIO/NA> | <DW/NA> | <G/NA> |
|----------------------------|-------------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------|-------------|---------|--------|
| IG902-H-LQA8               | High-config | China                           | LTE CAT4; LTE-FDD B1/B3/B5/B8;<br>LTE-TDD B34/B38/B39/B40/B41;<br>TD-SCDMA B34/B39; WCDMA<br>B1/B8; CDMA BCO; GSM<br>900/1800MHz | RS232×1 +<br>RS485×1 | IO          | NA      | NA     |
| IG902-H-LQA8-IO-DW-G       | High-config | China                           | LTE CAT4; LTE-FDD B1/B3/B5/B8;<br>LTE-TDD B34/B38/B39/B40/B41;<br>TD-SCDMA B34/B39; WCDMA<br>B1/B8; CDMA BCO; GSM<br>900/1800MHz | RS232×1 +<br>RS485×1 | IO          | DW      | G      |
| IG902-H-FQ58               | High-config | Europe &<br>APAC                | LTE CAT4; LTE-FDD<br>B1/B2/B3/B5/B7/B8/B20; LTE-TDD<br>B38/B40/B41; UMTS B1/B5/B8;<br>GSM B3/B8                                  | RS232×1 +<br>RS485×1 | IO          | NA      | NA     |
| IG902-H-FQ58-IO-DW-G       | High-config | Europe &<br>APAC                | LTE CAT4; LTE-FDD<br>B1/B2/B3/B5/B7/B8/B20; LTE-TDD<br>B38/B40/B41; UMTS B1/B5/B8;<br>GSM B3/B8                                  | RS232×1 +<br>RS485×1 | IO          | DW      | G      |
| IG902-H-FQ58-D485-RIO-DW-G | High-config | Europe &<br>APAC                | LTE CAT4; LTE-FDD<br>B1/B2/B3/B5/B7/B8/B20; LTE-TDD<br>B38/B40/B41; UMTS B1/B5/B8;<br>GSM B3/B8                                  | RS485×2              | RIO         | DW      | G      |
| IG902-H-FS39-IO            | High-config | North<br>America                | LTE CAT6; LTE-FDD<br>B2/B4/B5/B13/B17; UMTS B2/B5                                                                                | RS232×1 +<br>RS485×1 | IO          | NA      | NA     |
| IG902-H-FS39-IO-DW-G       | High-config | North<br>America                | LTE CAT6; LTE-FDD<br>B2/B4/B5/B13/B17; UMTS B2/B5                                                                                | RS232×1 +<br>RS485×1 | IO          | DW      | G      |
| IG902-H-FQ78               | High-config | Australia &<br>South<br>America | LTE CAT4; LTE-FDD<br>B1/B2/B3/B4/B5/B7/B8/B28; LTE-<br>TDD B40; UMTS B1/B2/B5/B8;<br>EDGE/GPRS/GSM<br>850/900/1800/1900MHz       | RS232×1 +<br>RS485×1 | NA          | NA      | NA     |
| IG902-H-FQ78-IO-DW-G       | High-config | Australia &<br>South<br>America | LTE CAT4; LTE-FDD<br>B1/B2/B3/B4/B5/B7/B8/B28; LTE-<br>TDD B40; UMTS B1/B2/B5/B8;<br>EDGE/GPRS/GSM<br>850/900/1800/1900MHz       | RS232×1 +<br>RS485×1 | IO          | DW      | G      |
| IG902-H-FQ88               | High-config | Japan                           | LTE CAT4; LTE-FDD<br>B1/B3/B8/B18/B19/B26; LTE-TDD<br>B41; WCDMA B1/B6/B8/B19                                                    | RS232×1 +<br>RS485×1 | IO          | NA      | NA     |
| IG902-H-FQ88-IO-DW-G       | High-config | Japan                           | LTE CAT4; LTE-FDD<br>B1/B3/B8/B18/B19/B26; LTE-TDD<br>B41; WCDMA B1/B6/B8/B19                                                    | RS232×1 +<br>RS485×1 | IO          | DW      | G      |
| IG902-H-FQ98               | High-config | South Korea                     | LTE CAT4; LTE-FDD<br>B1/B3/B5/B7/B8/B20; LTE-TDD<br>B38/B40/B41; WCDMA B1/B5/B8;<br>EDGE/GSM B3/B8                               | RS232×1 +<br>RS485×1 | IO          | NA      | NA     |

|                      |             |                      |                                                                                           |                   |    |    |    |
|----------------------|-------------|----------------------|-------------------------------------------------------------------------------------------|-------------------|----|----|----|
| IG902-H-FQ98-IO-DW-G | High-config | South Korea          | LTE CAT4; LTE-FDD B1/B3/B5/B7/B8/B20; LTE-TDD B38/B40/B41; WCDMA B1/B5/B8; EDGE/GSM B3/B8 | RS232×1 + RS485×1 | IO | DW | G  |
| IG902-H-EN00         | High-config | Global (No Cellular) | No 3G/4G module                                                                           | RS232×1 + RS485×1 | IO | NA | NA |
| IG902-H-EN00-IO-DW-G | High-config | Global (No Cellular) | No 3G/4G module                                                                           | RS232×1 + RS485×1 | IO | DW | G  |

## 6. Contact Us

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